# Temporal Caption/Query Adaptation with Spatial OPD

Eight-case post-seal development readout | 14 October 2026

Scope. Four historically exposed HC cases and four historically exposed VidSTG cases. GT metrics were read after prediction and diagnostic seals. This is not a fresh confirmation cohort; teacher-training overlap is unknown.

Main readout. T improves aggregate m_vIoU and m_tIoU over F and Projection in both datasets, with a small incremental gain over Projection on VidSTG. S and TS retain lower dense-GT m_sIoU than F in this panel. Spatial supervision used a student-selected pseudo-reference, not an independent geometry teacher; this outcome does not establish a general OPD failure.

Optimization issue. The fixed temporal KL+L2 objective ended above its initial value in 7/8 cases. Initial-minus-final is negative when the objective rose. Spatial loss is re-sampled on-policy, not a fixed-target convergence trace. No tuning or GT step selection followed.

## Dataset-level arm averages

| Set | Arm | vIoU % | tIoU % | dense sIoU % | d-vIoU vs F | d-tIoU vs F | d-sIoU vs F |
|---|---|---|---|---|---|---|---|
| HC | F | 22.21 | 46.41 | 47.89 | +0.00 | +0.00 | +0.00 |
| HC | Projection | 27.85 | 58.43 | 47.89 | +5.64 | +12.02 | +0.00 |
| HC | T | 31.29 | 60.49 | 47.89 | +9.08 | +14.07 | +0.00 |
| HC | S | 21.21 | 46.41 | 47.67 | -1.00 | +0.00 | -0.22 |
| HC | TS | 29.87 | 60.49 | 47.67 | +7.66 | +14.07 | -0.22 |
| VidSTG | F | 14.49 | 27.78 | 48.46 | +0.00 | +0.00 | +0.00 |
| VidSTG | Projection | 22.02 | 40.81 | 48.46 | +7.53 | +13.03 | +0.00 |
| VidSTG | T | 22.44 | 40.90 | 48.46 | +7.95 | +13.13 | +0.00 |
| VidSTG | S | 13.51 | 27.78 | 45.67 | -0.98 | +0.00 | -2.78 |
| VidSTG | TS | 21.14 | 40.90 | 45.67 | +6.65 | +13.13 | -2.78 |

T0 preserves Projection. Temporal adaptation has 512 active parameters and five fixed steps; spatial OPD has 1,792 active parameters and five actual steps. The final step is reported, not a GT-selected checkpoint. Per-case metrics and all six-step scalars are in the adjacent JSON exports.

## Per-case contrasts (percentage points)

| Case | T-F v/t (pp) | T-Projection v/t (pp) | S-F sIoU (pp) | TS-F sIoU (pp) |
|---|---|---|---|---|
| HC01 | -10.04 / -25.23 | -8.51 / -24.70 | +4.87 | +4.87 |
| HC02 | +16.45 / +24.73 | +12.32 / +21.28 | -7.94 | -7.94 |
| HC03 | +11.10 / +29.30 | +0.00 / +0.00 | +1.71 | +1.71 |
| HC04 | +18.80 / +27.49 | +9.92 / +11.63 | +0.48 | +0.48 |
| V01 | +15.55 / +23.90 | +2.45 / +3.70 | -3.89 | -3.89 |
| V02 | +15.01 / +25.46 | -0.44 / -0.94 | -1.05 | -1.05 |
| V03 | +0.88 / +1.63 | +0.00 / +0.00 | -1.97 | -1.97 |
| V04 | +0.36 / +1.52 | -0.31 / -2.37 | -4.23 | -4.23 |

# Per-case outcomes

All 20 case-by-arm rows are shown for each dataset. Values are percentages; positive and negative tails are preserved.

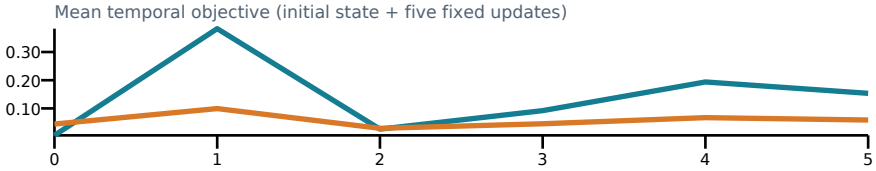
| HC / Case | Arm | vIoU | tIoU | dense sIoU |
|---|---|---|---|---|
| HC01 | F | 16.96 | 40.69 | 34.73 |
| HC01 | Projection | 15.42 | 40.16 | 34.73 |
| HC01 | T | 6.92 | 15.46 | 34.73 |
| HC01 | S | 16.69 | 40.69 | 39.60 |
| HC01 | TS | 6.58 | 15.46 | 39.60 |
| HC02 | F | 35.82 | 58.25 | 64.91 |
| HC02 | Projection | 39.94 | 61.70 | 64.91 |
| HC02 | T | 52.26 | 82.98 | 64.91 |
| HC02 | S | 30.93 | 58.25 | 56.97 |
| HC02 | TS | 45.37 | 82.98 | 56.97 |
| HC03 | F | 12.68 | 35.63 | 36.74 |
| HC03 | Projection | 23.78 | 64.94 | 36.74 |
| HC03 | T | 23.78 | 64.94 | 36.74 |
| HC03 | S | 13.21 | 35.63 | 38.45 |
| HC03 | TS | 24.89 | 64.94 | 38.45 |
| HC04 | F | 23.39 | 51.08 | 55.19 |
| HC04 | Projection | 32.26 | 66.94 | 55.19 |
| HC04 | T | 42.18 | 78.57 | 55.19 |
| HC04 | S | 24.03 | 51.08 | 55.67 |
| HC04 | TS | 42.67 | 78.57 | 55.67 |

| VidSTG / Case | Arm | vIoU | tIoU | dense sIoU |
|---|---|---|---|---|
| V01 | F | 36.18 | 54.62 | 63.49 |
| V01 | Projection | 49.28 | 74.81 | 63.49 |
| V01 | T | 51.73 | 78.52 | 63.49 |
| V01 | S | 33.95 | 54.62 | 59.60 |
| V01 | TS | 48.56 | 78.52 | 59.60 |
| V02 | F | 9.47 | 16.09 | 59.01 |
| V02 | Projection | 24.91 | 42.49 | 59.01 |
| V02 | T | 24.48 | 41.55 | 59.01 |
| V02 | S | 9.31 | 16.09 | 57.96 |
| V02 | TS | 24.06 | 41.55 | 57.96 |
| V03 | F | 6.01 | 14.04 | 48.26 |
| V03 | Projection | 6.89 | 15.67 | 48.26 |
| V03 | T | 6.89 | 15.67 | 48.26 |
| V03 | S | 5.76 | 14.04 | 46.29 |
| V03 | TS | 6.61 | 15.67 | 46.29 |
| V04 | F | 6.30 | 26.36 | 23.07 |
| V04 | Projection | 6.98 | 30.25 | 23.07 |
| V04 | T | 6.67 | 27.88 | 23.07 |
| V04 | S | 5.03 | 26.36 | 18.84 |
| V04 | TS | 5.32 | 27.88 | 18.84 |

# Optimization path and diagnostics

The fixed KL+L2 temporal objective finished above its initial value in 7/8 cases; negative initial-minus-final means it rose. Spatial loss is resampled on-policy. All six snapshots per case remain in JSON.

| Case | Initial loss | Final loss | Initial - final |
|---|---|---|---|
| HC01 | 0.00132 | 0.07399 | -0.07267 |
| HC02 | 0.01018 | 0.39776 | -0.38758 |
| HC03 | 0.00469 | 0.02505 | -0.02036 |
| HC04 | 0.00154 | 0.11724 | -0.11570 |
| V01 | 0.16789 | 0.12076 | +0.04713 |
| V02 | 0.00001 | 0.05849 | -0.05848 |
| V03 | 0.01017 | 0.03991 | -0.02973 |
| V04 | 0.00011 | 0.01619 | -0.01609 |




HC temporal | VidSTG temporal

Gradient comparison and within-query candidate correlations. QA/SSL comparisons use endpoint-NLL gradients on 512 frozen hidden-state parameters, not a vIoU derivative; GT gradients were not used for updates. Correlations are descriptive across four sources and 16 candidate rows per dataset.

| Case | QA-GT S0 | SSL-GT S0 | QA-GT native | SSL-GT native |
|---|---|---|---|---|
| HC01 | +0.047 | -0.337 | +0.337 | -0.103 |
| HC02 | -0.764 | +0.163 | -0.069 | +0.869 |
| HC03 | -0.329 | +0.400 | -0.307 | +0.388 |
| HC04 | -0.059 | +0.583 | -0.374 | +0.518 |
| V01 | +0.872 | +0.870 | +0.905 | +0.926 |
| V02 | -0.536 | +0.412 | -0.716 | +0.413 |
| V03 | -0.005 | -0.004 | +0.234 | -0.291 |
| V04 | +0.092 | -0.024 | +0.109 | -0.012 |

| Set | Sources | Rows | Reward-tIoU within-query r | Gain-tIoU within-query r |
|---|---|---|---|---|
| HC | 4 | 16 | -0.604 | +0.317 |
| VidSTG | 4 | 16 | +0.392 | -0.142 |

## Visual gain and fixed reward-shuffle readback

| Set | Mean G (nats/token) | Positive G / candidates | Mean QA-GT cosine | QA T - shuffled T (pp) |
|---|---|---|---|---|
| HC | 0.5222 | 16/16 | -0.2763 | +9.339 |
| VidSTG | 0.6166 | 16/16 | +0.1059 | +4.800 |

The SSL replay is a 512-parameter replay control, not the historical 66,306-parameter/32-panel fit. Higher visual-information gain G does not imply higher tIoU.

# Measured cost and evidence limits

| Stage | Calls / cases | Input / output tokens | Seconds | Peak GPU memory (GiB) |
|---|---|---|---|---|
| Teacher method | 64 | 43724 / 1268 | 31.410 | 15.88 |
| Auxiliary teacher diagnostics | 96 | 66165 / 467 | 26.950 | 15.92 |
| Student readout | 8 cases; 40 T + 40 S steps | -- | 56.394 | 5.35 |

The 96 auxiliary teacher requests comprise 64 paired-condition visual forced-likelihood requests, 8 reverse/replacement likelihood checks, 12 counterfactual caption calls, and 12 alignment calls. Eight reward-shuffle fits are CPU-only and add no teacher calls or full-model forwards. The student stage performed 8 temporal and 8 spatial fits with 40 optimizer steps in each branch.

## Counterfactual description controls (hashes only)

| Case | Control | Reward delta |
|---|---|---|
| HC01 | no_video | -0.438 |
| HC01 | replacement | -0.086 |
| HC01 | reversed | -0.354 |
| HC02 | no_video | -0.187 |
| HC02 | replacement | +0.239 |
| HC02 | reversed | +0.093 |
| V01 | no_video | -0.017 |
| V01 | replacement | -0.023 |
| V01 | reversed | +1.130 |
| V02 | no_video | -1.480 |
| V02 | replacement | -1.389 |
| V02 | reversed | -0.010 |

Only hashes and reward deltas are included; description and query text are omitted.

## Evidence boundaries

Historically exposed eight-case development panel; not fresh confirmation. Teacher-training overlap unknown. Ground-truth metrics were read after prediction seals; this was post-seal evaluation, not a no-GT run. Spatial reference was a student-selected pseudo-reference, not independent teacher geometry. QA/SSL gradients are endpoint-NLL gradients, not discrete vIoU derivatives; no GT gradient update. Visual-information gain and candidate correlations do not establish higher tIoU or causality. Negative spatial outcomes are retained but do not prove a universal OPD failure; no method promoted.

Root receipt: 8 cases, 184 metric readouts; official-metric reconstruction maximum absolute error 1.22e-15. Sanitized provenance and source hashes are in the JSON package.